

CBSE Class 7 Summative Assessment - 1 (SA1)

NCERT-Tutor v3 Master Exam Preparation Guide & Question Bank

Pedagogical Alignment: AlKosh [NCERT-TUTOR-v3](#) Multi-Stage DPO Framework (Science, Social, Computers, GK)

Standard: NCERT & State Board Class 7 Curriculum

Portion Scope:

- **Science:** Lesson 5 (*Changes Around Us: Physical and Chemical*) & Lesson 6 (*Adolescence: A Stage of Growth and Change*)
- **Social Studies:** Geography Ch 3 (*The Climates of India*), History Ch 8 (*The Age of Reorganisation*), Civics Ch 13 (*India's Tapestry of Communities*), Economics Ch 17 (*From Barter to Money*)
- **Computers:** Chapters 1 to 4 (*Excel Formulas & Functions, Charts, Advanced Features, Adobe Animate 2020*)
- **General Knowledge:** Unit 1 (*Fresh Finds: Union Budget 2025, State Governors, Sri Vijaya Puram, Robot Teachers*)

MODULE 1: GENERAL SCIENCE (LESSONS 5 & 6)



Core Scientific Foundations & NCERT Tutor Factuality

1. Lesson 5: Changes Around Us — Physical and Chemical Changes

- **Physical Change:**
 - A change in which only the physical properties (shape, size, state, appearance) of a substance change.
 - **No new substance is formed.** Most physical changes are reversible.
 - *Examples:* Melting of ice ($H_2O_{(s)} \rightarrow H_2O_{(l)}$), tearing of paper, dissolution of sugar in water, stretching of a rubber band.
- **Chemical Change (Chemical Reaction):**
 - A change in which one or more **new substances** with entirely distinct chemical identities and properties are formed.
 - Chemical changes are usually **irreversible** and accompanied by energy changes (heat/light absorption or release), gas evolution, or color change.
 - *Classic Reactions:*

1. Burning of Magnesium Ribbon:



When dissolved in water:



2. Copper Sulphate & Iron Nail Reaction (Displacement):



3. Rusting of Iron:



Essential Conditions: Both **Oxygen (O_2)** and **Moisture (H_2O)** are strictly required.

Prevention Methods: Painting, greasing, galvanisation (coating with Zinc), alloying (Stainless steel: Iron + Carbon + Cr, Ni, Mn).

4. **Crystallisation:** A physical purification process to obtain large, pure crystals of a substance from its saturated solution (e.g., pure Copper Sulphate crystals).

2. Lesson 6: Adolescence — A Stage of Growth and Change

• Adolescence & Puberty:

- Adolescence is the transitional developmental period between childhood and adulthood (typically ages 10–19).
- Puberty is the physiological milestone during which human reproductive systems mature.

• Major Physical Changes During Puberty:

- Sudden increase in height and skeletal growth.
- Development of secondary sexual characteristics: facial hair and deepening of voice in boys (enlargement of the larynx / Adam's Apple); breast development and commencement of menstruation in girls.
- Increased activity of sweat and sebaceous (oil) glands leading to acne.

• The Endocrine System & Master Hormones:

Endocrine Gland	Location	Primary Hormone	Physiological Function
Pituitary Gland	Base of Brain	Growth Hormone (GH) & TSH/FSH	Master Gland: Regulates body growth and controls all other endocrine glands
Thyroid Gland	Neck / Throat	Thyroxine	Regulates basal metabolic rate; deficiency causes Goitre
Pancreas	Behind Stomach	Insulin	Regulates blood glucose levels; deficiency causes Diabetes
Adrenal Glands	Atop Kidneys	Adrenaline	'Fight or Flight' hormone: controls heart rate, blood pressure, and stress response
Testes	Scrotum (Males)	Testosterone	Promotes male secondary sexual characteristics and sperm production
Ovaries	Pelvis (Females)	Estrogen & Progesterone	Promotes female secondary sexual characteristics and ovum maturation

• Nutritional Requirements of Adolescents:

- Balanced diet consisting of carbohydrates, proteins (for rapid cell growth), fats, vitamins, and minerals.

- **Iron** is crucial for hemoglobin synthesis and blood formation (sources: leafy green vegetables, jaggery, meat, citrus).
- **Calcium & Phosphorus** for bone density mineralization (sources: milk, curd, nuts).

Science High-Yield Question Drills (Worked Steps)

Question 1 (1 Mark — Objective / Assertion-Reason)

Assertion (A): Rusting of iron ships occurs much faster in seawater than in freshwater rivers.

Reason (R): Seawater contains high concentrations of dissolved salts, which accelerate the electrochemical process of rusting.

- **NCERT Tutor Explanation:** Both (A) and (R) are true, and (R) is the correct explanation of (A). The presence of electrolytes (salts) enhances electrical conductivity in water, significantly speeding up the oxidation of iron into hydrated ferric oxide.

Question 2 (2 Marks — Scientific Distinction)

Differentiate between Physical and Chemical Changes with two points each.

• **NCERT Worked Solution:**

1. **New Substance:** In a physical change, no new substance is formed (e.g. melting ice). In a chemical change, one or more new substances with completely new properties are formed (e.g. burning wood).
2. **Reversibility:** Physical changes are generally easily reversible by physical means. Chemical changes are typically irreversible.

Question 3 (3 Marks — Conceptual Breakdown)

Why is the pituitary gland referred to as the 'master gland'? Explain its role in triggering puberty.

• **NCERT Worked Solution:**

1. The pituitary gland is called the **master gland** because its secretions directly stimulate, coordinate, and regulate the endocrine secretions of all other target glands (thyroid, adrenals, testes, and ovaries).
2. At the onset of puberty, the pituitary releases gonadotropic hormones that instruct the testes to secrete **testosterone** in boys, and the ovaries to secrete **estrogen** in girls, initiating the secondary sexual characteristics.

Question 4 (5 Marks — Comprehensive Long Answer)

Explain the process of rusting of iron. Describe the chemical equation involved, the two essential conditions required, and three practical industrial methods used to prevent it.

• **NCERT Worked Solution:**

1. **Process:** Rusting is the slow oxidation of iron when exposed to atmospheric air and water vapor, forming a reddish-brown flaky coating called rust.
2. **Chemical Equation:** $4\text{Fe} + 3\text{O}_2 + 2x\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.
3. **Essential Conditions:** (i) Presence of Oxygen (O_2), (ii) Presence of Moisture / Water (H_2O). If either is absent, iron does not rust.
4. **Prevention Methods:**

- **Barrier Coating:** Applying paint, oil, or grease to prevent direct contact with air and moisture.
- **Galvanisation:** Depositing a protective layer of metallic Zinc on iron surfaces. Even if scratched, zinc oxidizes preferentially.
- **Alloying:** Combining iron with chromium and nickel to produce **stainless steel**, which is completely corrosion-resistant.

MODULE 2: SOCIAL STUDIES (GEOGRAPHY, HISTORY, CIVICS, ECONOMICS)



Core Humanities Foundations & NCERT Tutor Factuality

1. Geography Chapter 3: The Climates of India

- **Weather vs. Climate:**
- **Weather:** The short-term atmospheric state of a specific location at a specific moment (temperature, humidity, rainfall, wind velocity).
- **Climate:** The aggregate statistical average of weather patterns recorded over an extended period (typically 30–35 years).
- **Factors Influencing India's Climate:**
 1. **Latitude:** The Tropic of Cancer ($23^{\circ}30'N$) bisects India into a tropical southern half and a subtropical northern half.
 2. **Altitude:** Temperature decreases with elevation at the normal lapse rate ($6.5^{\circ}C$ per 1,000 m).
 3. **The Himalayan Barrier:** Protects India from frigid Siberian winds and traps rain-bearing monsoon winds.
 4. **Distance from Sea (Continentality):** Coastal regions experience equable climates; interior regions experience extreme temperatures.
 5. **Monsoon Winds:** The seasonal reversal of wind direction: Southwest Monsoon (June–September) and Retreating/Northeast Monsoon (October–November).

2. History Chapter 8: The Age of Reorganisation

- **Post-Gupta Transitions:**
- Fragmentation of centralized imperial rule led to the emergence of powerful regional dynasties (Harshavardhana, Chalukyas, Pallavas, Rashtrakutas, Cholas).
- Growth of **Samanta system:** Local feudal lords provided military troops to the king in exchange for land grants (*Brahmadeya* and *Devadana*).
- **Cultural & Temple Architecture:**
- Transition from rock-cut architecture to structural stone temples:
- **Dravida Style:** Towering *Gopurams*, stepped *Vimanas*, and enclosed pillared courtyards (Kanchipuram, Thanjavur).
- **Nagara Style:** Curvilinear *Shikharas* in northern and central India.

3. Civics Chapter 13: India's Tapestry of Communities / Cultural Coexistence

- **Unity in Diversity:**

- India's constitutional identity balances multi-linguistic, multi-religious, and multi-ethnic communities within a unified federal framework.

- **Constitutional Safeguards for Coexistence:**

- **Article 14:** Equality before the law.
 - **Article 15:** Prohibition of discrimination on grounds of religion, race, caste, sex, or place of birth.
 - **Secularism:** The state maintains principled distance and neutrality from all religions, according equal respect (*Sarva Dharma Sambhava*).
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4. Economics Chapter 17: From Barter to Money

- **The Barter System & Double Coincidence of Wants:**

- Direct exchange of commodities for commodities without the medium of money.
- *Critical Flaw:* Requires **double coincidence of wants**—both parties must desire precisely what the other offers.
- *Other Limitations:* Lack of a common unit of value, indivisibility of goods, difficulty in deferred payments and wealth storage.

- **Evolution of Money:**

- Commodity Money (Grain, cattle, shells) → Metallic Money (Gold, silver, copper coinage) → Paper Currency (Backed by sovereign guarantee) → Modern Banking (Cheques, Demand Drafts) → Digital Money (UPI, NEFT, RTGS).
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Social Studies High-Yield Question Drills (Worked Steps)

Question 1 (1 Mark — Objective)

Which parallel of latitude divides India into almost two equal halves, influencing its climatic zones?

- **NCERT Worked Answer:** The **Tropic of Cancer** (23°30'N).

Question 2 (2 Marks — Conceptual)

Explain the term 'Double Coincidence of Wants' with a real-life example.

- **NCERT Worked Solution:**

1. It is the condition where two economic agents simultaneously possess and desire to exchange the exact commodities the other wants.
2. *Example:* A farmer with surplus wheat who needs shoes must locate a shoemaker who not only has shoes but also specifically wants wheat in return. Without this mutual match, no barter transaction can occur.

Question 3 (3 Marks — Geography Analytical)

Why does Chennai receive most of its rainfall in the winter months (October–December), while Mumbai receives rainfall in July?

• **NCERT Worked Solution:**

1. Mumbai, located on the western coast, receives heavy rain from the **Southwest Monsoon (Arabian Sea branch)** during summer (June–August) as moisture-laden winds strike the Western Ghats.
2. Chennai, on the Coromandel coast, lies in the rain-shadow of the Western Ghats during the Southwest monsoon. However, during winter (October–November), the **Retreating / Northeast Monsoon winds** pick up moisture across the Bay of Bengal and deliver heavy rainfall to Tamil Nadu.

Question 4 (5 Marks — Economics & Civics Synthesis)

Explain why modern currency (paper notes and coins) is accepted as a medium of exchange even though it is not made of precious metals like gold or silver.

• **NCERT Worked Solution:**

1. **Sovereign Legal Guarantee:** Modern currency is authorized and backed by the Central Government. In India, the Reserve Bank of India (RBI) issues currency notes bearing the Governor's official promissory pledge.
2. **Legal Tender Status:** By law, no citizen or business within India can legally refuse payment made in Indian Rupees for settling financial obligations.
3. **Elimination of Barter Inefficiencies:** It acts as a universal measure and store of value, completely resolving the hurdle of double coincidence of wants.
4. **High Divisibility and Portability:** Unlike cattle or bulky grain, paper currency and digital tokens can be transported effortlessly and subdivided into fractional denominations.

MODULE 3: COMPUTERS & IT (CHAPTERS 1 TO 4)



Core Technical Principles & Syntax Guide

1. Chapter 1: Formulas and Functions in Excel 2016

• **Formulas vs. Functions:**

- **Formula:** A user-defined mathematical expression starting with `=` (e.g., `=A1+B1*2`).
- **Function:** A built-in, pre-programmed formula designed to perform specialized calculations (e.g., `=SUM(A1:A10)`).

• **Cell Referencing Rules:**

1. **Relative Referencing (`A1`):** Adjusts automatically when copied across rows or columns.
2. **Absolute Referencing (`$A$1`):** Fixed row and fixed column; does not change when copied.
3. **Mixed Referencing (`A$1` or `$A1`):** Either the column or the row is locked with `$`.

• **Standard Built-in Functions:**

- `=SUM(range)` : Computes the total sum.
- `=AVERAGE(range)` : Computes the arithmetic mean.
- `=MAX(range)` / `=MIN(range)` : Returns the highest / lowest numeric value.
- `=COUNT(range)` : Counts cells containing numeric entries.

2. Chapter 2 & 3: Charts and Advanced Features in Excel

- **Chart Components:** Chart Area, Plot Area, Data Series, Category Axis (*X*), Value Axis (*Y*), Legend, Data Labels, Gridlines.
 - **Chart Types:**
 - **Column / Bar Chart:** Compares discrete values across categories.
 - **Pie Chart:** Shows proportional contributions of parts to a whole (single data series).
 - **Line Chart:** Displays trends and changes over continuous time intervals.
 - **Sorting & Filtering:**
 - **Sorting:** Arranging records in ascending or descending sequence.
 - **Filtering (AutoFilter):** Selectively displaying records that fulfill specific criteria while temporarily hiding others.
 - **Conditional Formatting:** Automatically applying colors or styles to cells based on value thresholds.
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3. Chapter 4: Introduction to Adobe Animate 2020

- **Animation Workspace:**
 - **Stage:** The rectangular drawing canvas where visual graphics and animations appear.
 - **Timeline:** The panel coordinating frames and controlling the timing of animation sequences.
 - **Frames & Keyframes:**
 - *Frame:* A basic unit of time on the timeline.
 - *Keyframe (**F6**):* A critical frame where a major change, movement, or new drawing occurs.
 - *Blank Keyframe (**F7**):* An empty frame used to introduce a brand-new graphic.
 - **Tweening Techniques:**
 - **Shape Tween:** Morphs one vector shape into another over time.
 - **Classic / Motion Tween:** Smoothly moves an object, symbol, or group across the stage.
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Computers High-Yield Question Drills

Question 1 (2 Marks)

What will be the result of the Excel formula `=AVERAGE(10, 20, 30)` ? Write the formula to add cells A1 to A5.

• **Worked Answer:**

1. Average result: $\frac{10+20+30}{3} = 20$.
2. Sum formula: `"=SUM(A1:A5)"` .

Question 2 (3 Marks)

Differentiate between Relative and Absolute Cell Referencing in Excel with examples.

• **Worked Answer:**

- **Relative Referencing (**A1**):** Cell references adjust automatically relative to the destination cell when copied.
Example: `=A1+B1` copied down becomes `=A2+B2` .
- **Absolute Referencing (**\$A\$1**):** Cell coordinates are anchored with dollar signs (**\$**) and remain strictly unchanged regardless of where the formula is copied. Example: `=$A$1*B2` .

MODULE 4: GENERAL KNOWLEDGE (UNIT 1: FRESH FINDS)



Current Affairs & National Knowledge

1. Key Topics (Pages 5–9)

1. Union Budget 2025 (Page 5):

- Presented annually in Parliament by the Union Finance Minister on February 1st.
- Core allocations focus on capital infrastructure, green energy transition, agricultural modernization, and artificial intelligence technology hubs.

2. State Governors of India (Pages 6 & 7):

- The Governor acts as the constitutional head of a State, appointed by the President of India under Article 155.
- Holds office during the pleasure of the President.

3. Renaming of Port Blair to Sri Vijaya Puram (Page 8):

- In September 2024, the Government of India renamed Port Blair (capital of Andaman and Nicobar Islands) to **Sri Vijaya Puram**.
- *Significance:* Shedding colonial-era nomenclature to honor Netaji Subhash Chandra Bose's historic hoisting of the Indian Tricolour and the Andaman Islands' pivotal role in India's freedom struggle.

4. Robot Teachers in Schools (Page 9):

- Deployment of generative humanoid robot educators (such as **Iris** in Kerala schools) equipped with voice recognition, multi-lingual capabilities, and interactive educational content to assist primary education.



GK Quick Assessment Drill

1. Who appoints the Governor of an Indian State? → The President of India.
2. What was the historic capital city of the Andaman & Nicobar Islands renamed to? → Sri Vijaya Puram.
3. Name India's first AI robot teacher introduced in Kerala schools: → Iris.
4. On which date is the Union Budget customarily presented in Parliament? → February 1st.

SUMMARY OF SCORING PRINCIPLES FOR SA-1 EXAMS

1. **Science:** Always write chemical formulas with correct subscripts (FeSO_4 , CuSO_4) and label both the gland and its specific hormone.

2. **Social Studies:** Mention latitude coordinates ($23^{\circ}30'N$ for Tropic of Cancer) and use standard legal terms (*Double Coincidence of Wants*, *Article 14*, *Secularism*).
3. **Computers:** Start every Excel formula with an equals sign (=) and quote exact shortcut keys (F6 for Keyframe).
4. **General Knowledge:** Double-check exact current spelling (e.g. *Sri Vijaya Puram*).